

Ontology for `small.pdf` — VIT FFCS Academic Regulations 4.0

Hand-drawn ontology (revision 2), Step 1 of the KAG / hybrid-RAG classroom demonstration.

Source: `small.pdf` , 3 pages — the sole authority for what this graph may assert.

Source constraint: the PDF is truncated. Page 3 ends mid-sentence at "...Score is considered for admissions to the BDes". Everything after that point is absent from the source, including the M.Des admission route. The graph must not assert one.

Design principle: every node and edge below is traceable to a sentence in `small.pdf` . Where the document is silent, the ontology records the silence rather than guessing.

1. Entity types (node labels)

The labels are split into two groups. Core domain entities carry the academic and regulatory knowledge the demo reasons over; supporting entities carry provenance and vocabulary.

1a. Core domain entities

Label	Key properties	Instances found in the document
Regulation	name , version , applicable_from , incorporates_changes_till	v1.00, v1.10, v2.00, v2.10, v3.0, v3.1, v3.2, v4.0 , VIT Business School Regulations
Programme	name , level , duration_years , discipline , aliases , admission_information_status	B.Tech CSE / EEE / ECE / Civil / Mechanical, B.Tech Fashion Technology, M.Tech (2-yr), B.Des, M.Des
EntranceExam	name , full_name , mode , frequency , months , announced_via	VITEEE, VITMEE, UCEED, V-DAT
AdmissionCriterion	name , kind	VITEEE rank, VITMEE rank, 10+2 marks, subject of study, UCEED score, V-DAT score, work experience, merit list, counselling, seat availability
School	name , alias , functional_from	V-SIGN (VIT School of Design, July 2018), VIT Business School
Facility	name , detail	PROTICS Studio, 3D-iD Studio (30 iMacs), Smart PD Lab, Ergonomics Lab, Painting Booth
CouncilMeeting	number , date , date_known	18th, 20th, 27th, 28th, 37th, 46th, 59th, 71st, 72nd, Standing Committee (24 Sep 2010)
GoverningBody	name	Academic Council, Academic Policy Committee, Standing Committee of the AC, Management
Role	name	Vice-Chancellor, Proctor / Faculty Advisor, Student, Employer
University	name	VIT University
Campus	name , location	VIT Vellore ("preferred campus" implies others, none named)
CourseBasket	name , abbr	Foundation Core, DLE, Discipline Core, Discipline Elective, Specialization Elective, Open Elective, SEC, AEC, Project & Internship
System / Concept	name , introduced_year	FFCS (2008), VTOP, CAL, PBL, ICT, LTTPC, slot-based timetable
Feature	number , text	the 11 FFCS features, numbered 0-10
Cohort	name , academic_year , description	Students admitted 2021-22 onward

1b. Supporting / provenance entities

Label	Key properties	Instances
Document	id , title , pages , truncated	small.pdf (3 pages, truncated: true)
Chunk	id , page , text , embedding	~10-15 paragraph-level chunks, ids like chunk_p3_02
Abbreviation	short , expansion	the 17 on page 1: DLE, DE, SE, OE, SEC, AEC, DC, FC, LTPC, TH, LO, CO, FFCS, PBL, CAL, ICT, VTOP

channel is deliberately absent. See §2b.

Why separate core from supporting?

For a classroom KAG demonstration the split does three concrete jobs:

- 1. It makes the schema diagram readable.** MENTIONED_IN touches every node, so drawing it inline turns the diagram into a hairball. Splitting lets you show the domain layer on one slide and the provenance layer on another.
- 2. It matches the two retrieval paths.** Hybrid RAG enters through the supporting layer (vector search finds a chunk) and walks *up* into the core layer. KAG enters through the core layer (Cypher over domain entities) and walks *down* into the supporting layer to cite sources. Students see the two directions as two arrows on the same picture.
- 3. It keeps the LLM's text-to-Cypher prompt small.** When you generate Cypher from a natural language question you supply the schema; feeding only the core layer keeps the prompt short and stops the model from emitting MENTIONED_IN joins it does not need.

2. Relationship types

2a. Core relationships

Edge	Domain -> Range	Properties	Example instance
ADMITS_VIA	Programme -> EntranceExam	source_note	B.Tech CSE -> VITEEE; M.Tech -> VITMEE
EXEMPT_FROM	Programme -> EntranceExam	evidence	B.Tech Fashion Technology -> VITEEE
REQUIRES	Programme -> AdmissionCriterion	condition_group , mandatory	B.Des -> UCEED score (condition_group: "0R_1")
CONDUCTED_BY	EntranceExam -> University		VITEEE -> VIT University
OFFERS	School -> Programme		V-SIGN -> B.Des
HAS_FACILITY	School -> Facility		V-SIGN -> PROTICS Studio
LOCATED_AT	School -> Campus		V-SIGN -> VIT Vellore
HAS_CAMPUS	University -> Campus		VIT University -> VIT Vellore
APPROVED_IN	Regulation -> CouncilMeeting		v3.0 -> 37th AC meeting
CONVENED_BY	CouncilMeeting -> GoverningBody		37th meeting -> Academic Council
SUPERSEDES	Regulation -> Regulation		v4.0 -> v3.2 -> v3.1 -> ... -> v1.00
FOCUSES_ON	Regulation -> Concept		v3.0 -> CAL
APPLIES_TO	Regulation -> Cohort		v4.0 -> Students admitted 2021-22 onward
GOVERNED_BY	Regulation -> GoverningBody		v4.0 -> Academic Council
HEADED_BY	GoverningBody -> Role		Academic Policy Committee -> Vice-Chancellor
INTRODUCED_BY	System -> University	year	FFCS -> VIT University (2008)
HAS_FEATURE	System -> Feature		FFCS -> Feature 2 (slot-based timetable)
REFERS_TO	Feature -> CourseBasket		Feature 5 -> Open Elective
ADVISES	Role -> Role		Proctor / Faculty Advisor -> Student

`HAS_CAMPUS` is new in this revision: `Campus` is a core entity, so it needed a defined edge from `University` rather than only appearing as the target of `LOCATED_AT`.

2b. Supporting relationships

Edge	Domain -> Range	Example instance
ABBREVIATES	Abbreviation -> CourseBasket / System / Concept	OE -> Open Elective
MENTIONED_IN	any core entity -> Chunk	VITEEE -> chunk_p3_01
PART_OF	Chunk -> Document	chunk_p3_01 -> small.pdf

Removed: ANNOUNCED_VIA (Option A)

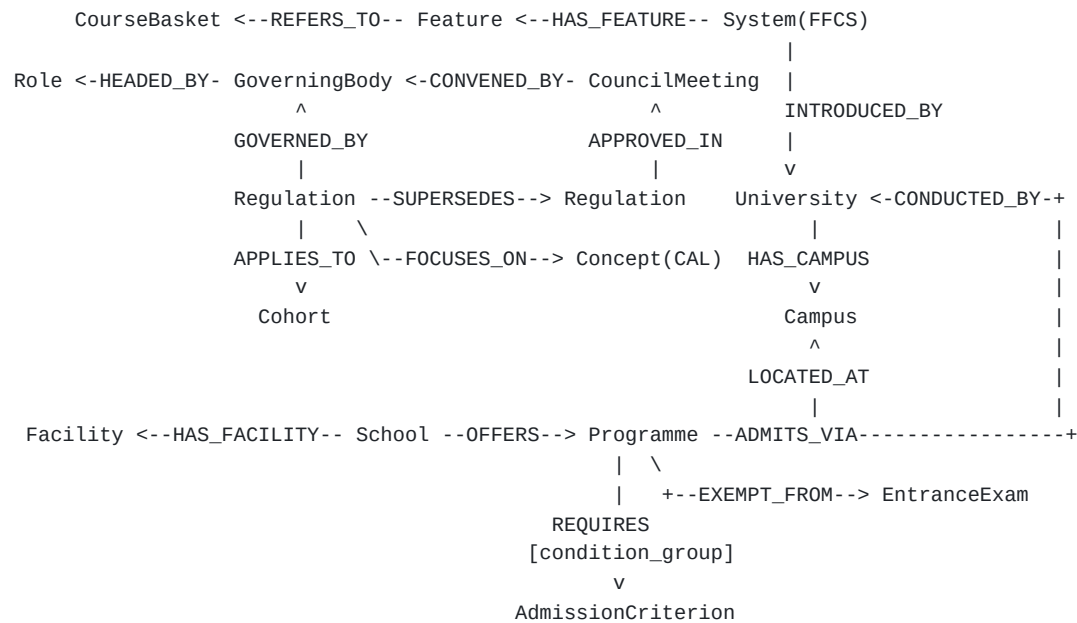
The previous revision had EntranceExam -> ANNOUNCED_VIA -> Channel with Channel never defined as a label. Rather than add a node type, the fact becomes a property:

```
(:EntranceExam {name: 'VITEEE', announced_via: ['media', 'university website']})
```

Two reasons this is the better teaching choice. First, "media / university website" is never reasoned *through* — no question in this demo hops from a channel to anything else, and an entity that is only ever a leaf is a property wearing a costume. Second, the source names the channels only as a vague pair, so Channel nodes would be two thin, ill-defined objects cluttering the diagram. Rule for the class: **promote a value to a node when you need to traverse it or attach facts to it; otherwise leave it a property.**

3. Graph shape

Core domain layer



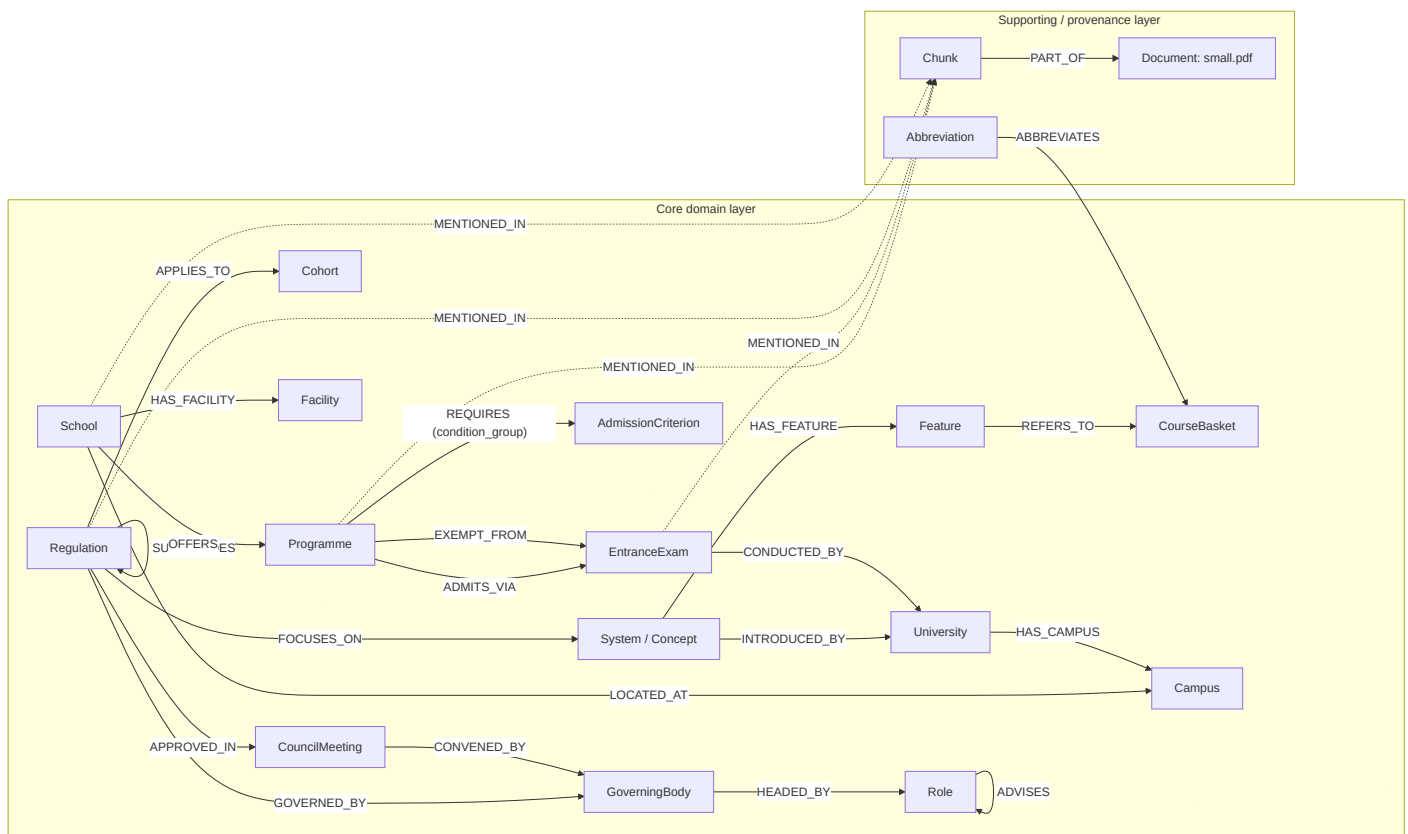
Supporting / provenance layer

```

every core entity --MENTIONED_IN--> Chunk --PART_OF--> Document(small.pdf)

Abbreviation --ABBREVIATES--> CourseBasket | System | Concept
  
```

4. Mermaid graph



5. Modelling decisions worth defending in the demo

5.1 EXEMPT_FROM is an explicit edge, not an absence

The source says of B.Tech Fashion Technology: **"There is NO Entrance Examination."** That is a positive assertion about a negative fact, and the graph stores it as one:

```

(:Programme {name: 'B.Tech Fashion Technology'})
-[:EXEMPT_FROM {evidence: 'There is NO Entrance Examination'}]->
(:EntranceExam {name: 'VITEEE'})
  
```

Why not simply omit ADMITS_VIA ? Because a missing edge is ambiguous, and in this very document the ambiguity is live. Both of these programmes lack an ADMITS_VIA edge:

Programme	Why the edge is missing	Correct answer to "which exam?"
B.Tech Fashion Technology	the document says there is none	"None — admission is by merit list"
M.Des	the document was cut off	"The document does not say"

Under the **open-world assumption** a graph database makes no claim about what it does not contain, so absence alone cannot distinguish these two. Reading absence as "no exam required" is a **closed-world** move, and here it would produce a confident wrong answer for M.Des. Storing the negation explicitly restores the distinction: Fashion Technology has an edge saying *no*, M.Des has no edge at all, and a status property (§5.3) says why.

This is the single most teachable idea in the ontology: **a knowledge graph can state that something is false; a vector index can only fail to retrieve it.**

5.2 REQUIRES carries a condition_group to preserve OR/AND structure

B.Des admission per the source: "a valid UCEED score **or** V-DAT (Design Aptitude Test) score **with** 10+2 marks in the qualifying examination."

That is (UCEED OR V-DAT) AND 10+2 . Three edges, with the alternatives tagged into a shared group:

```
(B.Des)-[:REQUIRES {condition_group: 'OR_1', mandatory: true}]->(UCEED score)
(B.Des)-[:REQUIRES {condition_group: 'OR_1', mandatory: true}]->(V-DAT score)
(B.Des)-[:REQUIRES {condition_group: 'AND', mandatory: true}]->(10+2 marks)
```

Reading rule: edges sharing a `condition_group` value beginning `OR_` are *alternatives* — satisfying any one satisfies the group. Edges in group `AND` are *individually required*. The overall requirement is the conjunction of all groups.

Flattening this to three independent mandatory edges would make the graph assert that a B.Des candidate needs UCEED **and** V-DAT **and** 10+2 marks, which is false and would wrongly reject a UCEED-only applicant. The property is the minimum machinery that keeps the logic honest; a full reified `Condition` node would be more general and, for a 3-page demo, more than it earns.

`mandatory: false` covers the one soft criterion in the source: for M.Tech, "work experience will be an added advantage" is modelled as (M.Tech)-[:REQUIRES {mandatory: false, note: 'added advantage'}]->(work experience) .

5.3 "Not stated" is recorded as a status, not left as silence

Every `Programme` carries:

```
admission_information_status ∈ { 'STATED', 'NOT_STATED_IN_SOURCE' }
```

M.Des gets `NOT_STATED_IN_SOURCE` with `status_reason: 'source document truncated mid-sentence on page 3'` . It gets **no** `REQUIRES` and **no** `ADMITS_VIA` edge — the graph does not invent UCEED or V-DAT for it.

Why a property rather than an `AdmissionRoute` node? An `AdmissionRoute` entity would be a node whose only job is to be absent or to say "unknown", which inverts the usual reason for creating a node. The property keeps the check to one predicate in Cypher, is uniform across all programmes, and reads naturally in a generated answer. Reach for a reified `AdmissionRoute` only if a later, fuller document gives routes their own attributes — effective dates, quotas, separate rule sets — which this one does not.

The payoff is that KAG can return a *correct negative*:

```
MATCH (p:Programme {name: 'M.Des'})
OPTIONAL MATCH (p)-[:ADMITS_VIA]->(e:EntranceExam)
OPTIONAL MATCH (p)-[:REQUIRES]->(c:AdmissionCriterion)
RETURN p.admission_information_status AS status, p.status_reason AS reason,
       collect(DISTINCT e.name) AS exams, collect(DISTINCT c.name) AS criteria
// -> status: 'NOT_STATED_IN_SOURCE', exams: [], criteria: []
```

which the answer layer renders as: *"The admission route for M.Des is not stated in the provided document."*

5.4 M.Tech is modelled as VITMEE; the source's wording is kept as provenance

The M.Tech paragraph introduces VITMEE as the M.Tech entrance exam, then says students are admitted "through counseling based on their **VITEEE** ranks" — a textual error in the source.

The domain layer represents the intended fact:

```
(M.Tech)-[:ADMITS_VIA {source_note: 'source paragraph inconsistently says "VITEEE ranks"; VITMEE is the M.Tech entrance exam'}->(VITMEE rank)
```

No `ADMITS_VIA` or `REQUIRES` edge to `VITEEE` or to `"VITEEE rank"` is created — propagating a typo into the graph would give a confidently wrong answer to "which exam do I sit for M.Tech?" The wording is preserved in the `source_note` and in the linked `chunk`, so the provenance trail still shows exactly what the document said. This is a good live illustration that **graph construction is an editorial act**, and that the edit must be recorded rather than hidden.

5.5 Unknown dates are flagged, not invented

The 72nd Academic Council meeting date appears in the source literally as `xxxxxxxxxxxxxxx` :

```
(:CouncilMeeting {number: 72, date: 'xxxxxxxxxxxxxxx', date_known: false})
```

The placeholder string is kept verbatim and the boolean makes it queryable, so a query can filter on `date_known` instead of parsing a nonsense date. v4.0 additionally carries `incorporates_changes_till`: `'71st Academic Council meeting'` as a property rather than a separate edge type — it is a scope note, not a relationship anything traverses.

6. Naming and alias notes

- Page 1 states that **"program" and "degree" are synonymous** and used interchangeably. `Programme` therefore carries an `aliases` list; the loader matches on normalised aliases so the two surface forms do not create two nodes.
- **V-SIGN = VIT School of Design** — one `School` node, `name`: `'VIT School of Design'`, `alias`: `'V-SIGN'`. Both surface forms appear in the text.
- **Degree surface forms normalise**: `BTech`, `B.Tech`, `B.Tech.` -> `B.Tech`; likewise `BDes` -> `B.Des`, `MTech` -> `M.Tech`, `MDes` -> `M.Des`.
- **Source spellings normalise**: `Mechanical` -> `Mechanical`; `abbreviations` -> `abbreviations`.
- **Exam acronyms expand** into `full_name`: `VITEEE` = VIT Engineering Entrance Exam, `VITMEE` = VIT Master's Entrance Exam, `UCEED` = Undergraduate Common Entrance Exam for Design, `V-DAT` = Design Aptitude Test.
- Loader rule: `MERGE` on the normalised canonical `name` only, never on a raw surface form. Every alias is stored on the single canonical node so entity linking from a chunk still resolves whichever form the text used.

7. Multi-hop questions and their graph paths

Question	Graph path	Expected answer
Which B.Tech programme needs no entrance exam?	(:Programme)-[:EXEMPT_FROM]->(:EntranceExam)	B.Tech Fashion Technology
Which Academic Council meeting approved the regulation that focused on CAL? (<i>control — single chunk, naive RAG also succeeds</i>)	(:Concept {name:'CAL'})<-[:FOCUSES_ON]-(:Regulation)-[:APPROVED_IN]->(:CouncilMeeting)	37th Academic Council meeting, 16 June 2015
What must a B.Des applicant provide?	(:Programme {name:'B.Des'})-[:r:REQUIRES]->(:AdmissionCriterion) grouped by r.condition_group	(UCEED or V-DAT) and 10+2 marks
How is a student admitted to M.Des?	no ADMITS_VIA / REQUIRES path; read admission_information_status	Not stated in the provided document
Which exam does an M.Tech applicant sit?	(:Programme {name:'M.Tech'})-[:ADMITS_VIA]->(:EntranceExam)	VITMEE
Which facilities belong to the school offering B.Des? (<i>genuinely split: chunk_p3_04 / chunk_p3_05</i>)	(:Programme {name:'B.Des'})<-[:OFFERS]-(:School)-[:HAS_FACILITY]->(:Facility)	PROTICS Studio, 3D-iD Studio, Smart PD Lab, Ergonomics Lab, Painting Booth
Which regulation version does v4.0 replace, and when was that one approved? (<i>genuinely split: chunk_p2_03 / chunk_p2_05</i>)	(:Regulation {version:'4.0'})-[:SUPERSEDES]->(:Regulation)-[:APPROVED_IN]->(:CouncilMeeting)	v3.2, approved at the 59th AC meeting, 24 Sep 2020
Which cohort does FFCS 4.0 apply to?	(:Regulation {version:'4.0'})-[:APPLIES_TO]->(:Cohort)	Students admitted 2021-22 onward
What are the course baskets and their abbreviations? (<i>genuinely split: chunk_p1_01 / chunk_p1_05</i>)	(:Abbreviation)-[:ABBREVIATES]->(:CourseBasket)	the eight baskets with DLE, DC, DE, SE, OE, SEC, AEC, FC
Where in the document does this come from?	(entity)-[:MENTIONED_IN]->(:Chunk)-[:PART_OF]->(:Document)	chunk id + page number for any answer above

8. What KAG can answer that naive vector RAG may get wrong

The three cases below are the demonstration. Each is a different *epistemic* situation, and the graph distinguishes all three while a vector index collapses them.

Measured baseline (step 4). Pure vector retrieval over the 18 chunks (BAAI/bge-small-en-v1.5 , cosine, top-3) puts the answer-bearing chunk in context for **6 of the 7 answerable questions**. Only the facilities question fails outright, missing chunk_p3_05 . This is a 3-page corpus, so retrieval is nearly solved and the honest claim is narrower than "the graph finds things vector search cannot": **the graph's advantage here is representation and reasoning, not recall**. Say that plainly in the session — it is the more durable lesson, and it survives a student pointing out that the whole document would fit in one prompt.

Case 1 — Explicit negative: Fashion Technology has no entrance examination

Question: "Which entrance exam do I need for B.Tech Fashion Technology?"

Measured on this corpus (step 4): retrieval **succeeds** — `chunk_p3_02`, which carries the negation, ranks 1st at 0.887. So this is not a retrieval failure. The risk is what sits beside it: rank 2 is `chunk_p3_01`, the paragraph stating that all B.Tech programmes require VITEEE, and rank 1 itself *opens* with "Dates of the entrance examinations such as VITEEE, VITMEE...". The LLM therefore receives the exemption and the VITEEE rule in one context window, with no structural marker saying which one governs Fashion Technology. Conflation is a live risk, but it is a reasoning failure, not a lookup failure.

KAG traverses `(:Programme {name:'B.Tech Fashion Technology'})-[:EXEMPT_FROM]->(:EntranceExam)` and returns the exemption as a fact, with the `evidence` property quoting the sentence. Nothing about VITEEE-for-other-programmes is in scope, so there is nothing to conflate.

Teaching point: embeddings encode topical similarity, not polarity. "X requires VITEEE" and "X requires no exam" sit close together in vector space and far apart in truth — so retrieval hands both to the model and leaves the disambiguation to it. The graph does the disambiguation *before* the model sees anything.

Case 2 — Explicit positive with logical structure: B.Des requires UCEED or V-DAT, plus 10+2

Question: "What do I need to get into B.Des?"

The right chunk is retrieved easily — this case does not fail at retrieval. It fails at *interpretation*. The LLM sees a run-on sentence spanning a line break ("valid UCEED score or V-DAT(Design Aptitude Test) score with 10+2 marks") and commonly flattens the disjunction, answering "you need UCEED, V-DAT and 10+2 marks" — which would wrongly turn away a UCEED-only applicant.

KAG returns three `REQUIRES` edges carrying `condition_group`, so the answer layer receives the structure explicitly and renders: "*Either a valid UCEED score or a V-DAT score, together with 10+2 marks.*"

Teaching point: the graph stores the *logical form*, not just the words. Boolean structure survives the trip; prose structure often does not.

Case 3 — Not stated: the M.Des admission route

Question: "How is a student admitted to M.Des?"

This is the sharpest case. Vector search retrieves the B.Des/UCEED paragraph, because it is the nearest thing in the document and it mentions design admissions. The chunk mentions M.Des a few lines earlier as a programme offered by V-SIGN. An LLM handed that context will very often answer "M.Des admission is based on a valid UCEED or V-DAT score" — a fluent, well-cited hallucination produced by the *absence* of contradicting text.

KAG finds no `ADMITS_VIA` and no `REQUIRES` edge, reads `admission_information_status = 'NOT_STATED_IN_SOURCE'`, and answers: "*The provided document does not state the admission route for M.Des.*"

Teaching point: naive RAG has no representation for "the corpus does not contain this". Retrieval always returns its top-k, and top-k is never empty. An explicit graph can return an empty result set — and an empty result set is an answer.

The three cases side by side

	Fashion Technology	B.Des	M.Des
Epistemic status	explicit negative	explicit positive, structured	not stated
In the graph	EXEMPT_FROM edge exists	3 REQUIRES edges + condition_group	no edges; status property
Correct answer	"No entrance exam"	"(UCEED or V-DAT) and 10+2"	"Document does not say"
Retrieval (measured, top-3)	PASS — rank 1	PASS — rank 1	returns B.Des chunk at 0.854
Likely naive-RAG failure	conflates with the adjacent VITEEE rule	flattens OR into AND	hallucinates UCEED
Root cause	negation is not encoded in embeddings	logical form lost in prose	no representation of absence

9. Why this ontology suits a KAG / hybrid-RAG demonstration

It is small enough to hold in your head. Fifteen core labels, nineteen core relationships, one 3-page source. Students can read the whole graph in the Neo4j browser in a single screen, which means every answer the system gives can be checked by eye against the graph — the thing that almost never happens with a realistic corpus.

It has genuine multi-hop structure — verified against the actual chunks. Step 2 chunked the source at 700 characters and checked which questions really do straddle a boundary. Three do: the facilities question (chunk_p3_04 / chunk_p3_05), abbreviation-to-basket (chunk_p1_01 / chunk_p1_05), and the version chain (chunk_p2_03 / chunk_p2_05). These are honest wins for the graph, not contrived ones.

The CAL question is **not** one of them: chunk_p2_05 happens to contain both "FFCS Regulations 3.0, which focussed on CAL" and "37th Academic Council meeting held on 16th June 2015", so naive RAG answers it correctly. It is kept in the question set deliberately, as a **control**. A demo in which the graph wins every single question invites the suspicion that the questions were picked to make it win; showing one where retrieval alone suffices makes the other results credible, and makes the real point — that the advantage depends on where the boundaries fall, which is a property of the chunking, not a law of nature.

Its three epistemic cases are naturally occurring. The explicit negative, the OR/AND requirement, and the truncation gap were not planted — they are in the source. That makes the comparison in §8 credible rather than staged.

The provenance layer makes both architectures runnable on one graph. Because every core entity links to a chunk , the same database serves both demos:

- *Hybrid RAG*: embed the question, vector-search chunk nodes, follow MENTIONED_IN backwards to the entities in those chunks, expand one hop, and hand the LLM chunk text **plus** graph triples.
- *KAG*: generate Cypher from the question, execute it against the core layer, then follow MENTIONED_IN forwards to attach the supporting sentences as citations, and present the traversal itself as the reasoning chain.

Running both over one graph isolates the variable — the retrieval strategy — so the difference students observe is attributable to the architecture and not to different data.

It models its own defects. date_known: false , the M.Tech source_note , truncated: true on the document, and NOT_STATED_IN_SOURCE on M.Des together demonstrate the discipline the whole exercise is meant to teach: a knowledge graph is only as trustworthy as its willingness to say what it does not know.